Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

# IT13: ADBMS

## Chapter 2:SQL

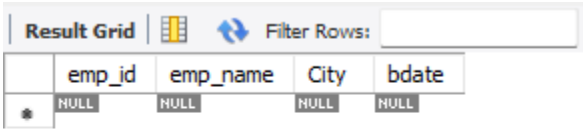
## Practical Assignment 1 Solution Template

**Write the SQL query and paste a screenshot of the output below the question.**

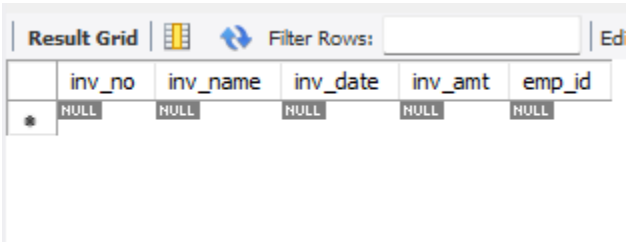
1. Create a database and two tables in it.

```
create database Information;
use Information;

create table Employee (emp_id int primary key, emp_name varchar(20), City varchar(20), bdate date);
select * from Employee;
```




```
create table Investor (inv_no int primary key, inv_name varchar(20), inv_date date, inv_amt int, emp_id int);
select * from Investor;
```




2. Insert at least 5 rows into the table.

```
insert into Employee values(1, 'Tanvi', 'Pune', '2004-08-25');

insert into Employee values(2, 'Shyam', 'Solapur', '2000-09-22');
```

insert into Employee values(3, 'Kaushik', 'Nagpur', '1990-09-19');

insert into Employee values(4, 'Drishti', 'Kolhapur', '1999-09-22');

insert into Employee values(5, 'Tarika', 'Solapur', '1998-07-12');

Result Grid | Filter Rows:

| emp_id | emp_name | City | bdate |
|---|---|---|---|
| 1 | Tanvi | Pune | 2004-08-25 |
| 2 | Shyam | Solapur | 2000-09-22 |
| 3 | Kaushik | Nagpur | 1990-09-19 |
| 4 | Drishti | Kolhapur | 1999-09-22 |
| 5 | Tarika | Solapur | 1998-07-12 |
| NULL | NULL | NULL | NULL |

Employee 23

insert into Investor (inv_no, inv_name, inv_date, inv_amt, emp_id) values

(1, 'Rohan', '1978-02-22', 99999, 2),

(2, 'Sita', '1974-07-02', 88888, 1),

(3, 'Gita', '1972-01-18', 77777, 5),

(4, 'Babita', '1966-02-28', 66666, 3),

(5, 'Sharon', '2008-11-22', 55555, 4);

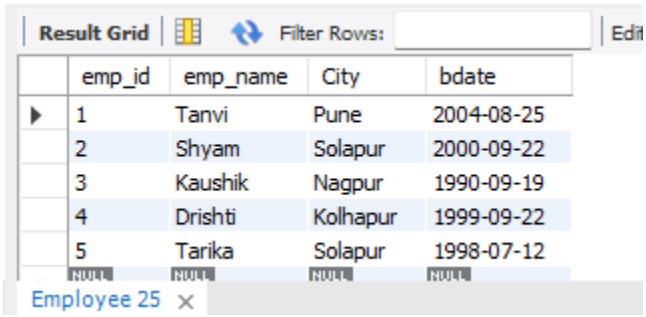
Result Grid | Filter Rows: | Edit:

| inv_no | inv_name | inv_date | inv_amt | emp_id |
|---|---|---|---|---|
| 1 | Rohan | 1978-02-22 | 99999 | 2 |
| 2 | Sita | 1974-07-02 | 88888 | 1 |
| 3 | Gita | 1972-01-18 | 77777 | 5 |
| 4 | Babita | 1966-02-28 | 66666 | 3 |
| 5 | Sharon | 2008-11-22 | 55555 | 4 |
| NULL | NULL | NULL | NULL | NULL |

Investor 24

3. Display all the information in the Employee table.

select * from Employee;



Result Grid | Filter Rows: | Edit

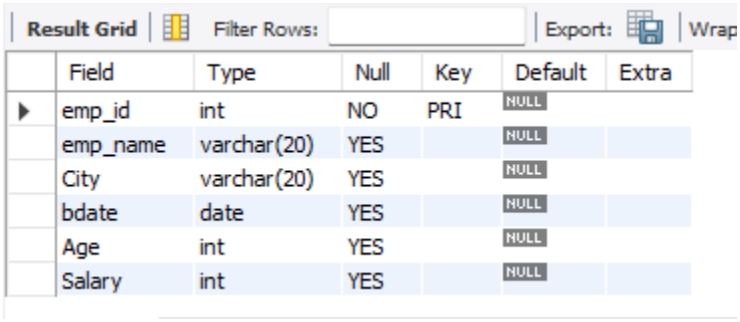
| emp_id | emp_name | City | bdate |
|---|---|---|---|
| 1 | Tanvi | Pune | 2004-08-25 |
| 2 | Shyam | Solapur | 2000-09-22 |
| 3 | Kaushik | Nagpur | 1990-09-19 |
| 4 | Drishti | Kolhapur | 1999-09-22 |
| 5 | Tarika | Solapur | 1998-07-12 |
| NULL | NULL | NULL | NULL |

Employee 25

4. Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).

alter table Employee add column(Age int, Salary int);

desc Employee;



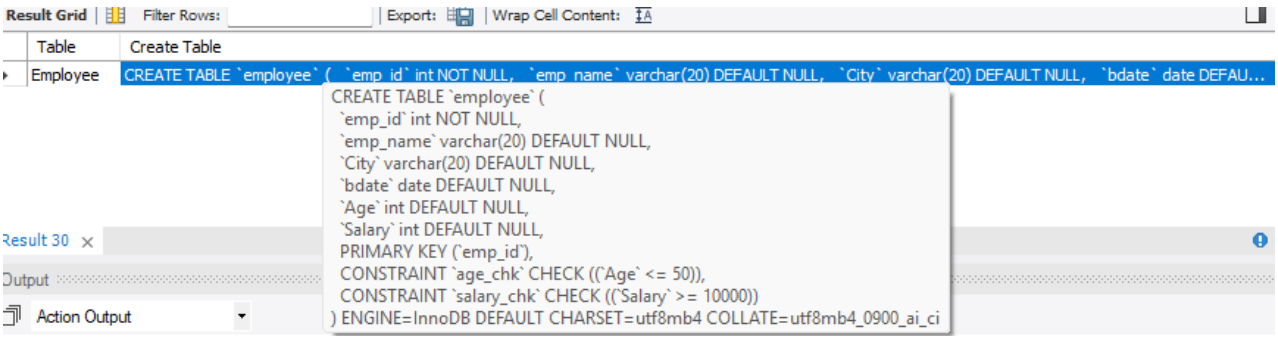
Result Grid | Filter Rows: | Export: | Wrap

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| emp_id | int | NO | PRI | NULL | |
| emp_name | varchar(20) | YES | | NULL | |
| City | varchar(20) | YES | | NULL | |
| bdate | date | YES | | NULL | |
| Age | int | YES | | NULL | |
| Salary | int | YES | | NULL | |

alter table Employee add constraint age_chk check (Age <= 50);

alter table Employee add constraint salary_chk check (Salary >= 10000);

show create table Employee;



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Table | Create Table |
|---|---|
| Employee | CREATE TABLE `employee` ( `emp_id` int NOT NULL, `emp_name` varchar(20) DEFAULT NULL, `City` varchar(20) DEFAULT NULL, `bdate` date DEFAU... |

```
CREATE TABLE `employee` (
  `emp_id` int NOT NULL,
  `emp_name` varchar(20) DEFAULT NULL,
  `City` varchar(20) DEFAULT NULL,
  `bdate` date DEFAULT NULL,
  `Age` int DEFAULT NULL,
  `Salary` int DEFAULT NULL,
  PRIMARY KEY (`emp_id`),
  CONSTRAINT `age_chk` CHECK ((`Age` <= 50)),
  CONSTRAINT `salary_chk` CHECK ((`Salary` >= 10000))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
```

Result 30

Output

Action Output

5. Add a column 'Age' to the Investor table.

alter table Investor add column(Age int);

desc Investor;

Result Grid | Filter Rows: | Export: | Wra

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| inv_no | int | NO | PRI | NULL | |
| inv_name | varchar(20) | YES | | NULL | |
| inv_date | date | YES | | NULL | |
| inv_amt | int | YES | | NULL | |
| emp_id | int | YES | | NULL | |
| Age | int | YES | | NULL | |

6. Change the data type of inv_name from varchar to text in the Investor table

alter table Investor modify inv_name text;

desc Investor;

Result Grid | Filter Rows: | Export:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| inv_no | int | NO | PRI | NULL | |
| inv_name | text | YES | | NULL | |
| inv_date | date | YES | | NULL | |
| inv_amt | int | YES | | NULL | |
| emp_id | int | YES | | NULL | |
| Age | int | YES | | NULL | |

7. Change the name of the column/field Emp_id to E_id.

alter table Employee rename column emp_id to e_id;

desc Employee;

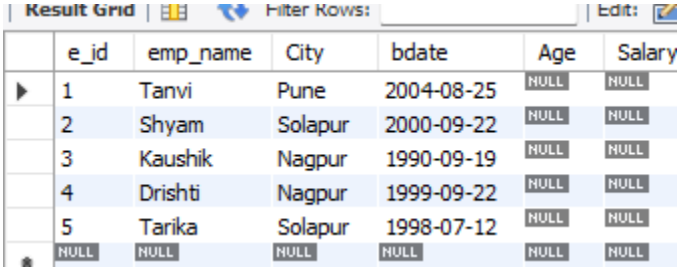
select * from Employee;

Result Grid | Filter Rows: | Export: | W

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| e_id | int | NO | PRI | NULL | |
| emp_name | varchar(20) | YES | | NULL | |
| City | varchar(20) | YES | | NULL | |
| bdate | date | YES | | NULL | |
| Age | int | YES | | NULL | |
| Salary | int | YES | | NULL | |

8. Update the city of Emp_id-4 to the current city, Nagpur.

update Employee set City = 'Nagpur' where e_id = 4;

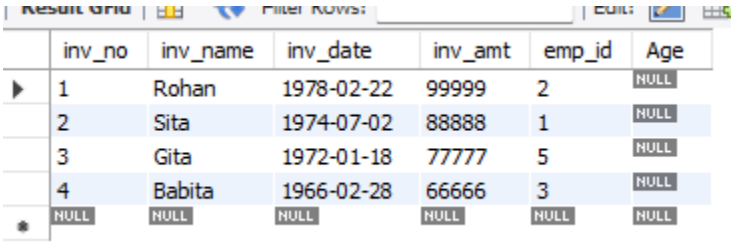


| e_id | emp_name | City | bdate | Age | Salary |
|---|---|---|---|---|---|
| 1 | Tanvi | Pune | 2004-08-25 | NULL | NULL |
| 2 | Shyam | Solapur | 2000-09-22 | NULL | NULL |
| 3 | Kaushik | Nagpur | 1990-09-19 | NULL | NULL |
| 4 | Drishti | Nagpur | 1999-09-22 | NULL | NULL |
| 5 | Tarika | Solapur | 1998-07-12 | NULL | NULL |
| NULL | NULL | NULL | NULL | NULL | NULL |

9. Delete one record from the Investor table.

```
delete from Investor where inv_no = 5;

select * from Investor;
```

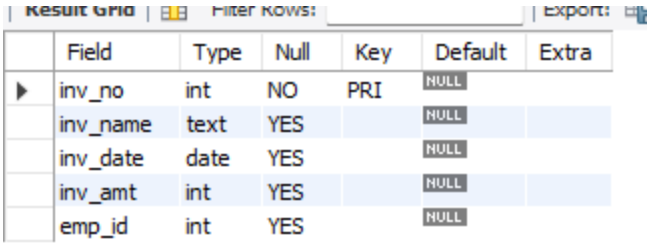


| inv_no | inv_name | inv_date | inv_amt | emp_id | Age |
|---|---|---|---|---|---|
| 1 | Rohan | 1978-02-22 | 99999 | 2 | NULL |
| 2 | Sita | 1974-07-02 | 88888 | 1 | NULL |
| 3 | Gita | 1972-01-18 | 77777 | 5 | NULL |
| 4 | Babita | 1966-02-28 | 66666 | 3 | NULL |
| NULL | NULL | NULL | NULL | NULL | NULL |

10. Drop the column 'Age' from the Investor table.

```
alter table Investor drop column Age;

desc Investor;
```

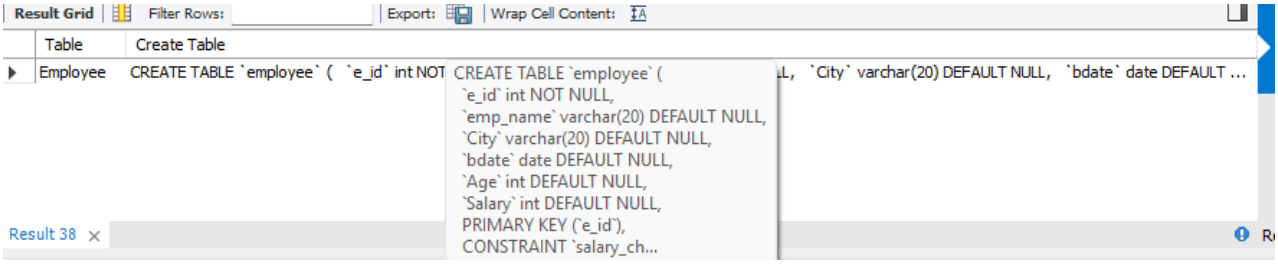


| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| inv_no | int | NO | PRI | NULL | |
| inv_name | text | YES | | NULL | |
| inv_date | date | YES | | NULL | |
| inv_amt | int | YES | | NULL | |
| emp_id | int | YES | | NULL | |

11. Drop the CHECK constraint on the 'Age' column from the Employee table.

```
alter table Employee drop constraint age_chk;

show create table Employee;
```



| Table | Create Table |
|---|---|
| Employee | CREATE TABLE `employee` ( `e_id` int NOT NULL, `emp_name` varchar(20) DEFAULT NULL, `City` varchar(20) DEFAULT NULL, `bdate` date DEFAULT NULL, `Age` int DEFAULT NULL, `Salary` int DEFAULT NULL, PRIMARY KEY (`e_id`), CONSTRAINT `salary_ch... |

Result 38

12. Create foreign key constraints between the Employee and Investor tables

```
alter table Investor add constraint inv_emp_fk foreign key Investor(emp_id) references
Employee(e_id);
desc Investor;
```

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| inv_no | int | NO | PRI | NULL | |
| inv_name | text | YES | | NULL | |
| inv_date | date | YES | | NULL | |
| inv_amt | int | YES | | NULL | |
| emp_id | int | YES | MUL | NULL | |

13. Display any 4 columns from the Investor table.

```
select inv_no, inv_name, inv_date, inv_amt from Investor;
```

| inv_no | inv_name | inv_date | inv_amt |
|---|---|---|---|
| 1 | Rohan | 1978-02-22 | 99999 |
| 2 | Sita | 1974-07-02 | 88888 |
| 3 | Gita | 1972-01-18 | 77777 |
| 4 | Babita | 1966-02-28 | 66666 |
| NULL | NULL | NULL | NULL |

14. Rename a table.

```
rename table Employee to Employee01;
desc Employee01;
select * from Employee01;
show create table Employee01;
```

| Table | Create Table |
|---|---|
| Employee01 | CREATE TABLE `employee01` ( `e_id` int NOT NULL, `emp_name` varchar(20) DEFAULT NULL, `City` varchar(20) DEFAULT NULL, `bdate` date DEFAU... |

15. Delete all the records using the TRUNCATE command.

truncate Investor;

truncate Employee;

16. Drop both tables from the database.

drop table Investor;

drop table Employee1;